Unit 6 Exponential & Logarithmic Functions Study Guide

Objectives…Remember that big colored sheet of paper you taped into your notebook at the beginning of the unit??

A. Basics of Exponential Functions

- [ ] I can evaluate and interpret exponential functions.
- [ ] I can write an exponential function (given two points, from a scenario).
- [ ] I can solve applications using exponential functions.
- [ ] I can use the constant *e* to represent continuous growth.

B. Logarithms

- [ ] I can evaluate logarithmic expressions.
- [ ] I can condense and expand expressions using logarithmic properties.
- [ ] I can explain the relationship between exponents and logarithms.

C. Solving Exponential & Logarithmic Equations (with Extraneous Solutions)

- [ ] I can solve exponential equations.
- [ ] I can solve logarithmic equations.
- [ ] I can identify extraneous solutions when solving exponential and logarithmic equations.

D. Graphing Logarithmic Functions

- [ ] I can graph a logarithmic function (parent function).
- [ ] I can state the domain and range of a logarithmic function.

Essential Questions

1. What is an exponential function?
2. What are some of the characteristics of the graph of an exponential function?
3. What is the natural base *e*?
4. What are some of the characteristics of the graph of a logarithmic function?
5. How can you transform the graphs of exponential and logarithmic functions?
6. How can you use properties of exponents to derive properties of logarithms?
7. How can you solve exponential and logarithmic equations?

Topics Covered

★ Lesson 1 – Exponential Functions
★ Lesson 2 – The Natural Base *e*
★ Lesson 3 – Logarithms
★ Lesson 4 – Transformations of Exponential & Logarithmic Functions
★ Lesson 5 – Properties of Logarithms
★ Lesson 6 – Solving Exponential & Logarithmic Functions
★ Lesson 7 – Modeling with Exponential & Logarithmic Functions

How can I practice/study for this assessment?

Your notes, practice, and homework creates a "practice guide"!

1. Complete any unfinished practice problems
2. Redo practice problems
3. DeltaMath
4. Visit our Google Classroom page for digital versions of in class activities!!
5. Visit our Google Classroom page for any additional digital versions of practice!!

Unit 6 Exponential & Logarithmic Functions Practice Guide

Complete the following problems in your HOMEWORK notebook.

1. **Short Answer Questions**
   a. Write down all equations/formulas from this unit that you need to know. Then, state everything you know about them.

   b. What is the relationship between a logarithmic function & an exponential function?

   c. State the definition of a logarithm. Be sure to include any limiting factors/restrictions.

   d. State the log properties of this unit and what they are.

   e. What is the change of base formula? When would you use it?

2. **You can use a scientific calculator for all of the following problems. *Remember to still show your work!***
   a. As Algebratown gets smaller, the population P(t) of its high school decreases by 12% each year. In the year 2015 (t = years since 2015) it had 125 students. What year starts with 75 students?

   b. Ms. Collins invests $10,000 into an account that earns 6.2% interest, compounded quarterly so she can save for a new car. How much money will she have in 4 years?

   c. I want to have $12,000 at the end of 5 years. I have $7500 to invest. If I want to find an account that compounds weekly, what must the rate be in order for me to achieve this?

   d. Decide if the following information represents an exponential function. If so, find the function and state if it is a growth or decay.

x	1	2	3	4
f(x)	1.64	2.13	2.77	3.60

3. **You can use a scientific calculator for all of the following problems. *Remember to still show your work!***

   The table shows the number of rabbits *r* in a particular forest *t* years after a orest fire. Write and use an exponential model to find how many years it will take for the rabbit population to surpass 20,000.

Years after the fire, *t*	0	1	2	3	4	5
Rabbits, *r*	20	60	180	540	1620	4860

4. **You can use a scientific calculator for all of the following problems.** ***Remember to still show your work!***

The table shows the values (in thousands) of the average salary of NBA players x years after 1990. Write and use an exponential model to find the year when the average NBA player's salary will be more than $4 million.

Years (after 1990), *x*	1	2	3	4	5	6	7
Salary (thousands), *y*	25	50	100	200	400	800	1600

Name ____ Date _________

6.7 Practice B

In Exercises 1 and 2, determine the type of function represented by the table. Explain your reasoning.

1.

x	0	2	4	6	8
y	$\frac{1}{8}$	$\frac{1}{2}$	2	8	32

2.

x	0	1	2	3
y	8	12	18	27

In Exercises 3–8, write an exponential function $y = ab^x$ whose graph passes through the given points.

3. $(1, 10), (2, 20)$

4. $(1, 18), (3, 162)$

5. $(2, 36), (3, 72)$

6. $(3, 375), (4, 1875)$

7. $(2, 3.6), (5, 777.6)$

8. $(2, 8), (5, 512)$

9. Describe and correct the error in determining the type of function represented by the data.

✗

x	0	2	4	6	8
y	−2	−4	−8	−16	−32

$\times 2$ $\times 2$ $\times 2$ $\times 2$

The outputs have a common ratio of 2, but the outputs are negative, so the data does not represent a recognizable function.

In Exercises 10 and 11, determine whether the data show an exponential relationship. Then write a function that models the data.

10.

x	1	3	5	7	9
y	64	32	16	8	4

11.

x	0	10	20	30	40
y	0	15	30	45	60

12. Use a graphing calculator to find an exponential model for the data in the table.

x	2	5	6	8	9
y	7.65	25.819	38.728	87.138	130.71

06 - Exponential & Logarithmic Equations - Study Guide (H)

1. What is the value of $\log_2 8$?

2. What is the value of $\log_5 125$?

3. What is the value of $\ln \frac{1}{e^3}$?

4. What is the value of $\log \sqrt[4]{1000}$?

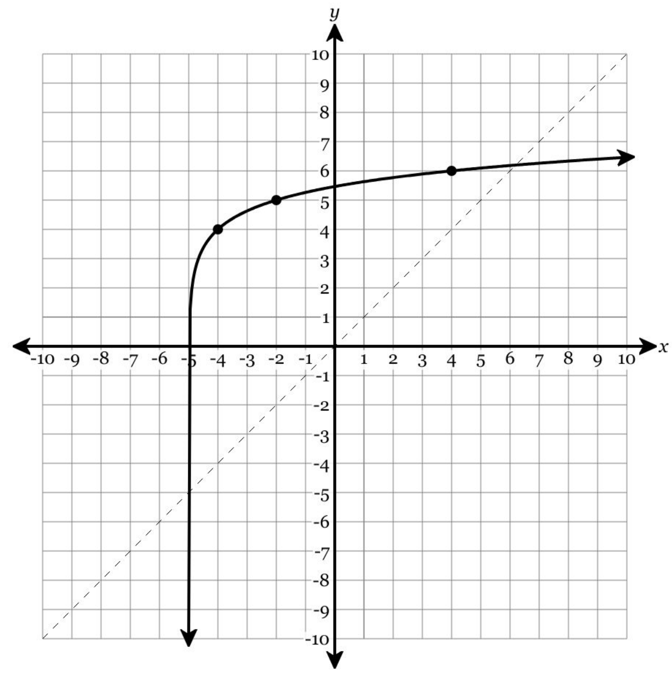
5. What is the value of $\log_3 81$?

6. Evaluate: $\log_{64} 128$

7. Evaluate: $\log_{128} \frac{1}{4}$

8. Evaluate: $\log_{243} 9$

9. The function $f(x) = \log_3(x+5) + 4$ is graphed below. Plot all lattice points of the inverse. Use the labeled points as your guide.



The inverse can be given by the function

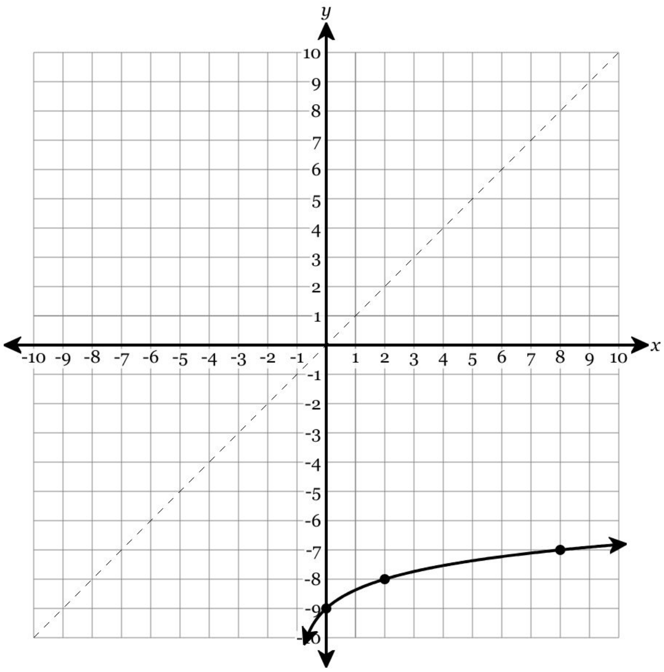
$f^{-1}(x) =$ ______________________________.

It has a (vertical / horizontal) asymptote of ____________.

The range of the inverse function is __________, and it is

(increasing / decreasing) on its domain of __________.

10. The function $f(x) = \log_3(x+1) - 9$ is graphed below. Plot all lattice points of the inverse. Use the labeled points as your guide.



The inverse can be given by the function

$f^{-1}(x) =$ ______________________.

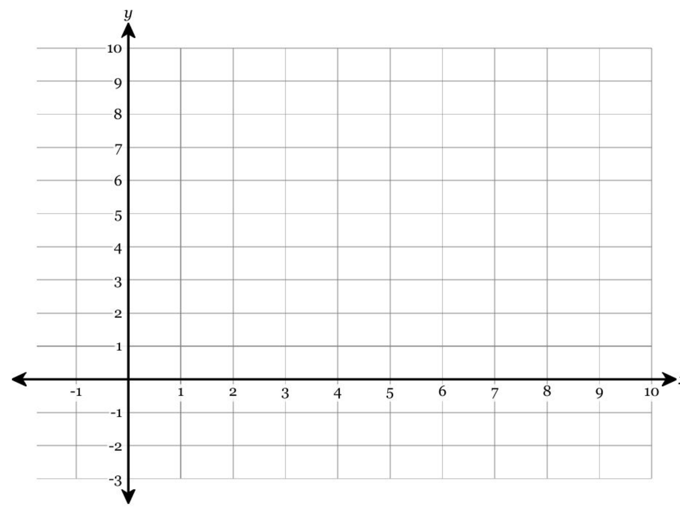
It has a (vertical / horizontal) asymptote of __________.

The range of the inverse function is __________, and it is

(increasing / decreasing) on its domain of __________.

11. Given the function $y = \log_2 x + 4$, fill in the table of values, use the table of values to graph the function, and then identify the function's domain and range.

x	$\log_2 x$	$y = \log_2 x + 4$
$\frac{1}{4}$		
$\frac{1}{2}$		
1		
2		
4		
8		

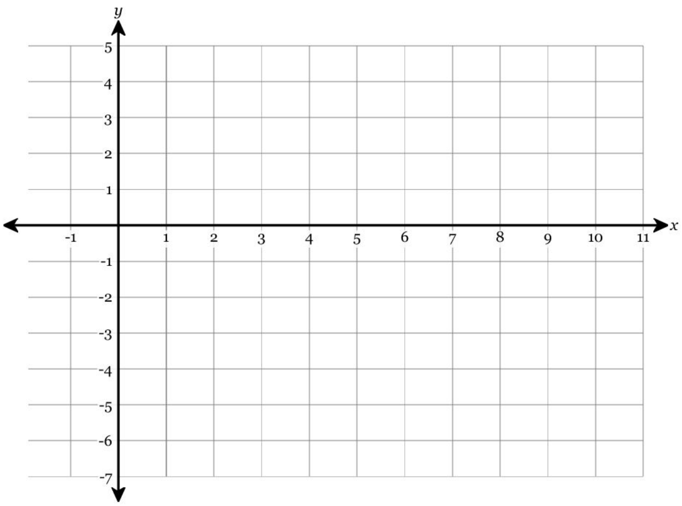


Domain:

Range:

12. Given the function $y = \log_3 x - 1$, fill in the table of values, use the table of values to graph the function, and then identify the function's domain and range.

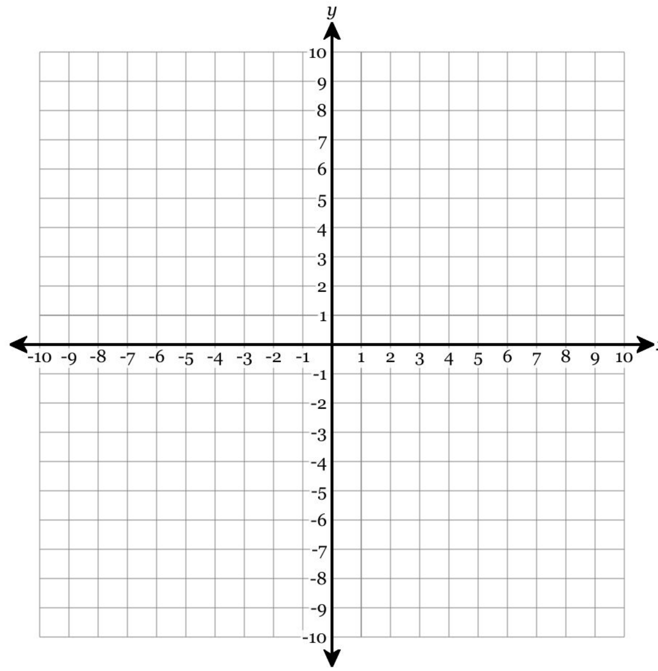
x	$\log_3 x$	$y = \log_3 x - 1$
$\frac{1}{9}$		
$\frac{1}{3}$		
1		
3		
9		



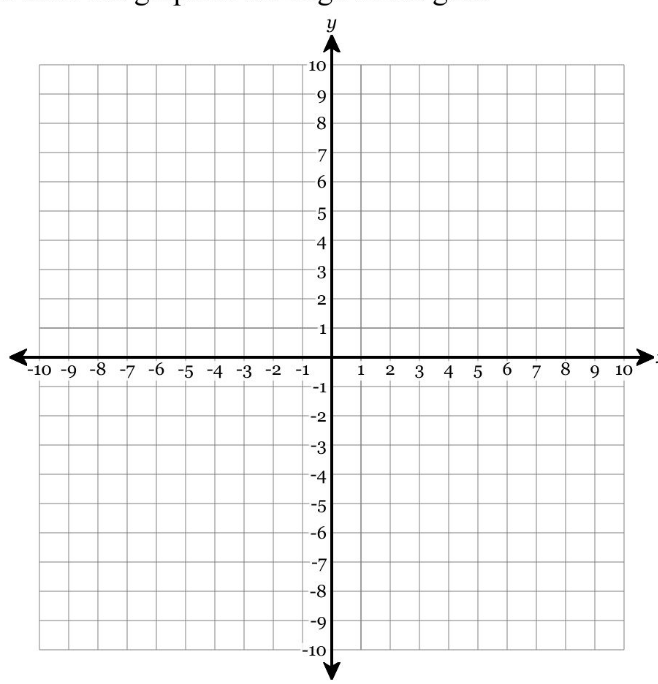
Domain:

Range:

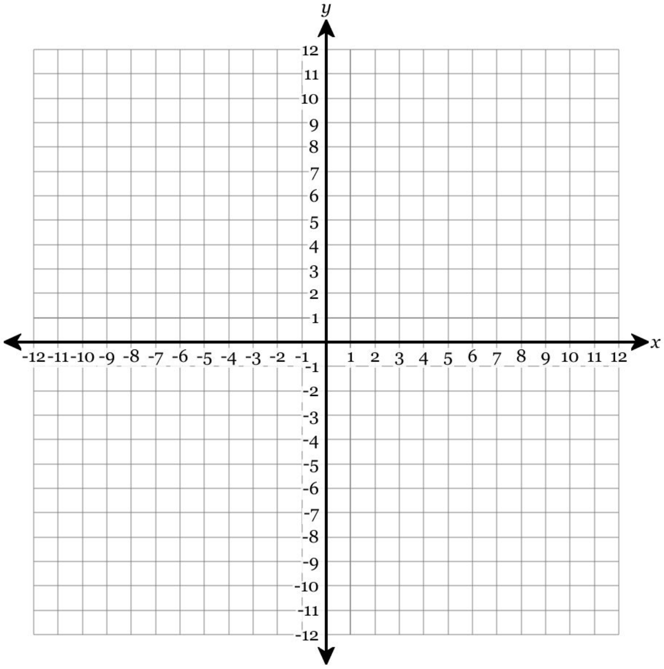
13. Use technology to find points and then graph the function $y = -2\log_2(x+4)+3$. Plot *at least four* points on the axes below. Then draw a function through all of your points. Make sure that you draw the graph to the edge of the grid.



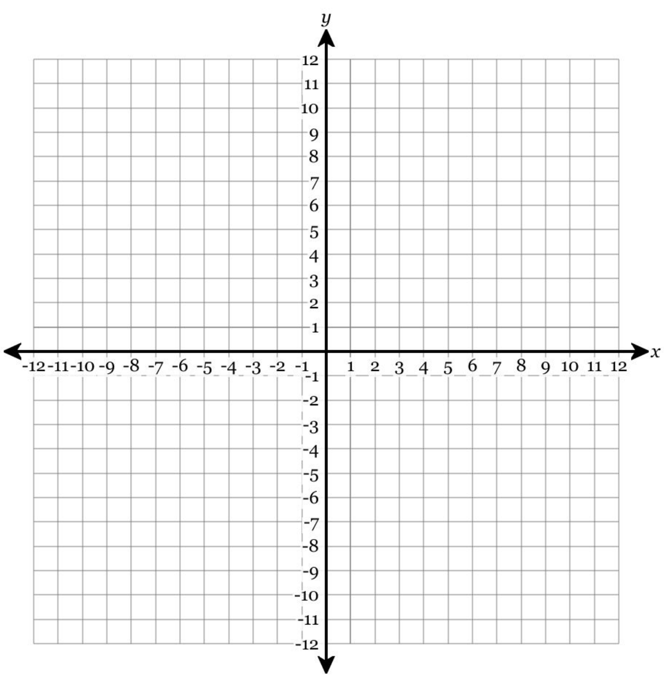
14. Use technology to find points and then graph the function $y = \log_2(-x+8)-5$. Plot *at least five* points with integer coordinates on the axes below. Then draw a function through all of your points. Make sure that you draw the graph to the edge of the grid.



15. Graph the function $f(x) = 3\log_5(-x+3) - 8$ on the axes below. You must plot the asymptote and any two points with integer coordinates.



16. Graph the function $f(x) = -2\left(\frac{1}{3}\right)^{x+4} - 4$ on the axes below. You must plot the asymptote and any two points with integer coordinates.



17. Expand the logarithm fully using the properties of logs. Express the final answer in terms of $\log x$, and $\log y$.

$$\log x^2 y^5$$

18. Expand the logarithm fully using the properties of logs. Express the final answer in terms of $\log x$, and $\log y$.

$$\log \frac{x^3}{y^4}$$

19. Expand the logarithm fully using the properties of logs. Express the final answer in terms of $\log x$, $\log y$, and $\log z$.

$$\log \frac{z^3}{x^5 \sqrt[3]{y}}$$

20. Expand the logarithm fully using the properties of logs. Express the final answer in terms of $\log x$, $\log y$, and $\log z$.

$$\log \frac{z^4}{x^2 y^4}$$

21. Expand the logarithm fully using the properties of logs. Express the final answer in terms of $\log x$, $\log y$, and $\log z$.

$$\log \frac{y^2 z}{\sqrt[3]{x}}$$

22. Find the numerical value of the log expression.

$$\log a = 9 \qquad \log b = -12 \qquad \log c = -4$$

$$\log \frac{\sqrt{c}}{a^4 b^4}$$

23. Find the numerical value of the log expression.

$$\log a = 6 \qquad \log b = -6 \qquad \log c = -9$$

$$\log \frac{a^2 c^2}{\sqrt{b}}$$

24. Condense the logarithm:

$$3 \log x - 3 \log r$$

25. Condense the logarithm:

$$\log q + g \log y$$

26. Condense the logarithm:

$$8 \log y + \log r$$

27. Write the expression below as a single logarithm in simplest form.

$$\log_b 8 + \log_b 3$$

28. Write the expression below as a single logarithm in simplest form.

$$3 \log_b 4 - \log_b 8$$

29. Write the expression below as a single logarithm in simplest form.

$$\log_b 2 - \log_b 2$$

30. Solve for x, rounding to the *nearest hundredth*.

$$2 \cdot 2^{\frac{x}{5}} = 1$$

31. Solve for x, rounding to the *nearest hundredth*.

$$700e^x = 70$$

32. A person invests 8000 dollars in a bank. The bank pays 5.25% interest compounded semi-annually. To the *nearest tenth of a year*, how long must the person leave the money in the bank until it reaches 14800 dollars?

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

33. Element X decays radioactively with a half life of 12 minutes. If there are 150 grams of Element X, how long, *to the nearest tenth of a minute*, would it take the element to decay to 2 grams?

$$y = a(.5)^{\frac{t}{h}}$$

34. A person invests 6000 dollars in a bank. The bank pays 4.25% interest compounded monthly. To the *nearest tenth of a year*, how long must the person leave the money in the bank until it reaches 8000 dollars?

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

35. Solve for x *to the nearest 10th.*

$$300 = 4990(0.95)^x - 150$$

36. Solve for x *to the nearest 10th.*

$$810 = 900(1.06)^{\frac{x}{8}-2} - 490$$

37. A town has a population of 5000 and grows at 5% every year. To the nearest tenth of a year, how long will it be until the population will reach 12300?

38. An element with a mass of 900 grams decays by 7% per minute. To the nearest tenth of a minute, how long will it be until there are 230 grams of the element remaining?

39. Solve for all values of x

$$\log_{-2x-6}(3x^2 + 7x - 16) = 2$$

40. Solve for all values of x

$$\log_5(x^2 - 3x + 7) = 2$$

41. Solve for all values of x

$$\log_{x+4}(2x^2 + 9x + 14) = 2$$

42. Solve for all values of x

$$\log_{x+7}(3x^2 + 4x + 1) = 2$$

43. Solve for all values of x

$$\log_{x+8}(3x^2 + 2x + 4) = 2$$

44. Solve for the *exact* value of x.

$$6\ln(4x - 10) - 2 = 40$$

45. Solve for the *exact* value of x.

$$3\ln(5x - 1) + 17 = 29$$

46. Solve for the *exact* value of x.

$$5\ln(5x - 4) - 15 = 15$$

47. Express the given expression as an integer or as a fraction in simplest form.

$$\log_8\left(8^{-\frac{1}{2}}\right)$$

48. Express the given expression as an integer or as a fraction in simplest form.

$$\log\left(\frac{1}{\sqrt[3]{10^4}}\right)$$

49. Write the exponential equation as a logarithmic equation. You do not need to solve for x.

$$e^2 = 4x - 8$$

50. Write the exponential equation as a logarithmic equation. You do not need to solve for x.

$$8^{3x} = 7$$

51. Write the log equation as an exponential equation. You do not need to solve for x.

$$\log_3(4x) = 2$$

52. Write the log equation as an exponential equation. You do not need to solve for x.

$$\log_{(x+5)}(8) = 3$$

53. Solve for a positive value of x.

$$\log_4(256) = x$$

54. Solve for a positive value of x.

$$\log_3(x) = 5$$

55. Solve the following logarithm problem for the positive solution for x.

$$\log_{10} x = 2$$

56. Solve the following logarithm problem for the positive solution for x.

$$\log_x 4 = 2$$

57. Solve the following logarithm problem for the positive solution for x.

$$\log_3 x = 3$$

58. Solve for the exact value of x. Evaluate radicals wherever possible. Leave answers as whole numbers, fractions, or radicals (if they cannot be evaluated). No exponents or decimals.

$$-2\log_3(x) + 7 = 3$$

59. Solve for the exact value of x. Evaluate radicals wherever possible. Leave answers as whole numbers, fractions, or radicals (if they cannot be evaluated). No exponents or decimals.

$$9\log_8(x) - 11 = -5$$

60. Solve for the exact value of x. Evaluate radicals wherever possible. Leave answers as whole numbers, fractions, or radicals (if they cannot be evaluated). No exponents or decimals.

$$-9\log_2(x) + 15 = 15$$

61. Solve for the exact value of x.

$$\log_3(2x) + \log_3(6) = 3$$

62. Solve for the exact value of x.

$$\log_2(8x) - \log_2(7) = 4$$

63. Solve for the exact value of x.

$$\log_7(2x) + \log_7(2) = 5$$

64. Solve for all values of x.

$$\log_3(3x-4) + \log_3(x-2) = 0$$

65. Solve for all values of x.

$$\log_3(x^2 - 4) - \log_3(x + 2) = 3$$

66. Solve for all values of x.

$$\log_5(x - 5) + \log_5(x - 10) = \log_5(x + 2)$$

67. Consider the functions f and g given by $f(x) = \log_2(x + 9) - \log_2(x - 9)$ and $g(x) = \log_2(x - 1)$. Find the x-coordinates of all points of intersection.

Scan the QR code or visit deltamath.com/qr/6PY5R-282TR to view full solutions.